

Probabilistic World Models for Autonomous Driving

"The Brain in the Box"

The Goal:

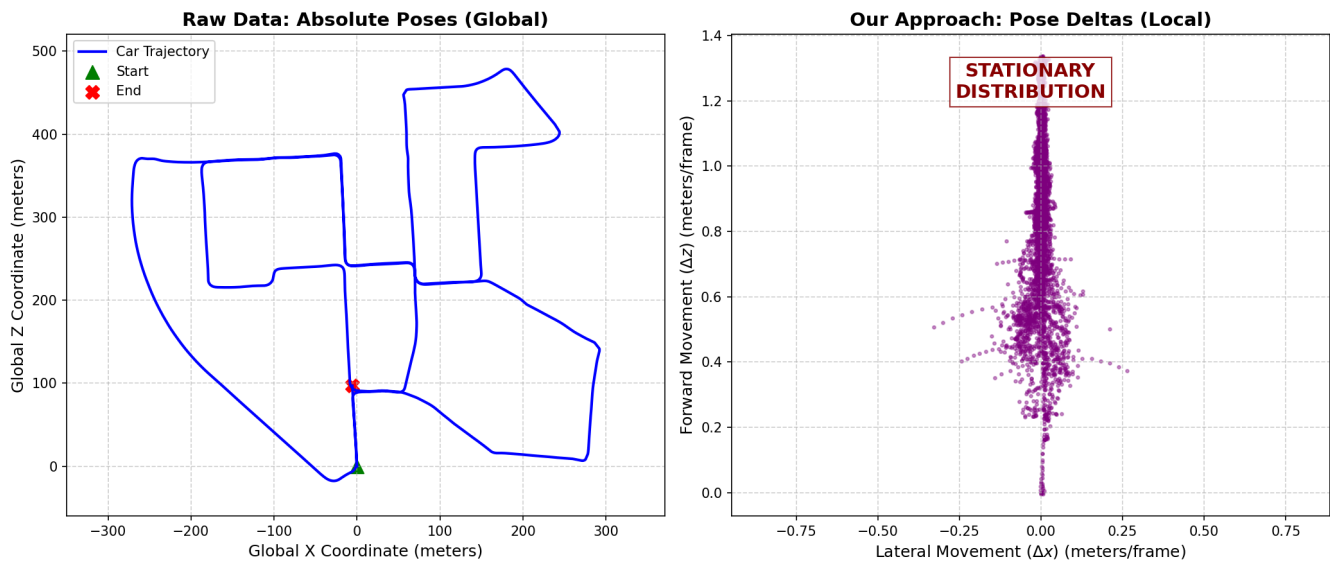
Don't just teach the car to drive.

Teach the car to simulate reality.

1. The Data Insight: How to Move

Problem: Absolute coordinates vary wildly.

Solution: Learn **Pose Deltas** (Velocity).



We learn the physics of motion, not specific maps.

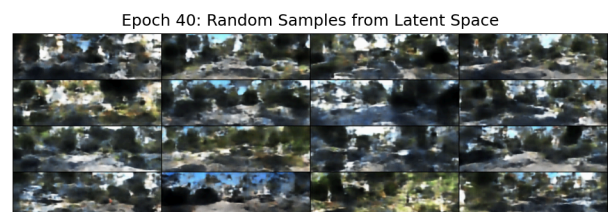
2. Vision Model (VAE)

The "Eyes": Compressing images into features (z).

Reconstruction (Real Input)



Random Sampling (Dreaming)



Reconstruction (Real Input)

Captures structure.

Random Sampling (Dreaming)

Blurry/Invalid states.

Insight: The latent space contains "holes" (undefined regions).

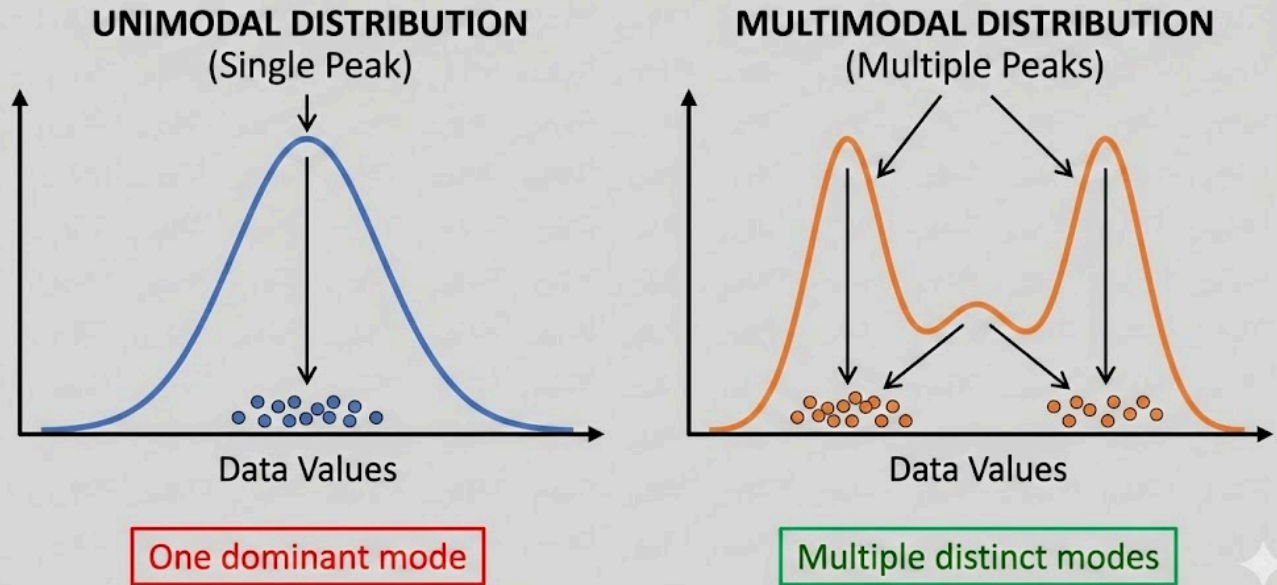
3. The Memory Model (MDN-RNN)

The "Brain": Predicting uncertain futures.

Why Probabilistic? The "Fork in the Road"

Deterministic models average choices → Crash.

MULTIMODAL CONCEPT IN DATA DISTRIBUTIONS



4. The Architecture

Combining Memory + Uncertainty + Movement

graph TD

Zt["Visual Input (z_t)"] --> LSTM[LSTM Core (Context/Memory)]

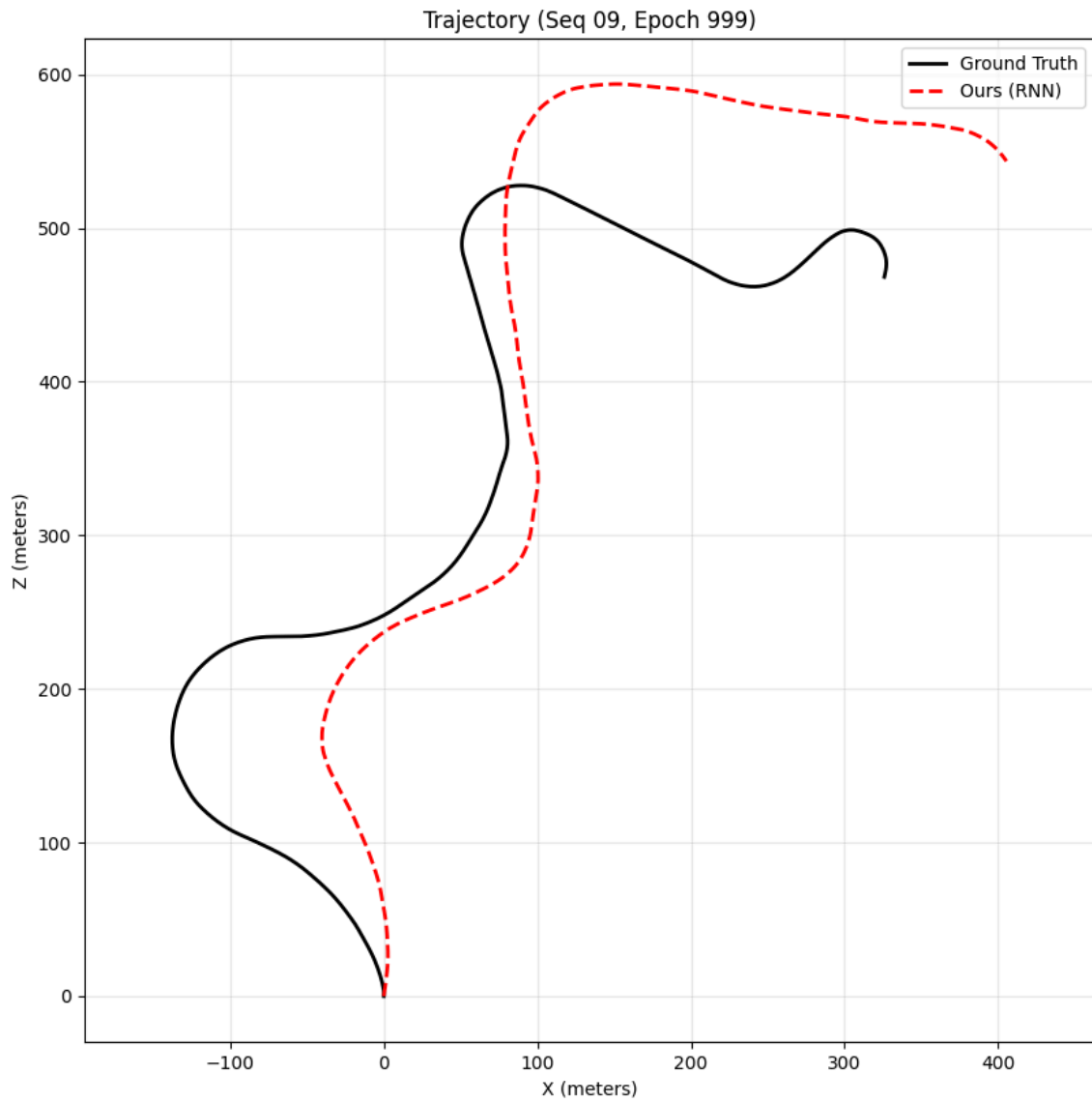
LSTM --> GMM["VISION HEAD : Probabilistic Future : $P(z_{t+1} | z_t)$ "]

LSTM --> Pose["POSE HEAD : Movement Delta : (Δx , Δy , $\Delta \theta \dots$)"]

5. Results: Trajectory Prediction

Open-Loop Testing

(Model sees ground truth at every step)

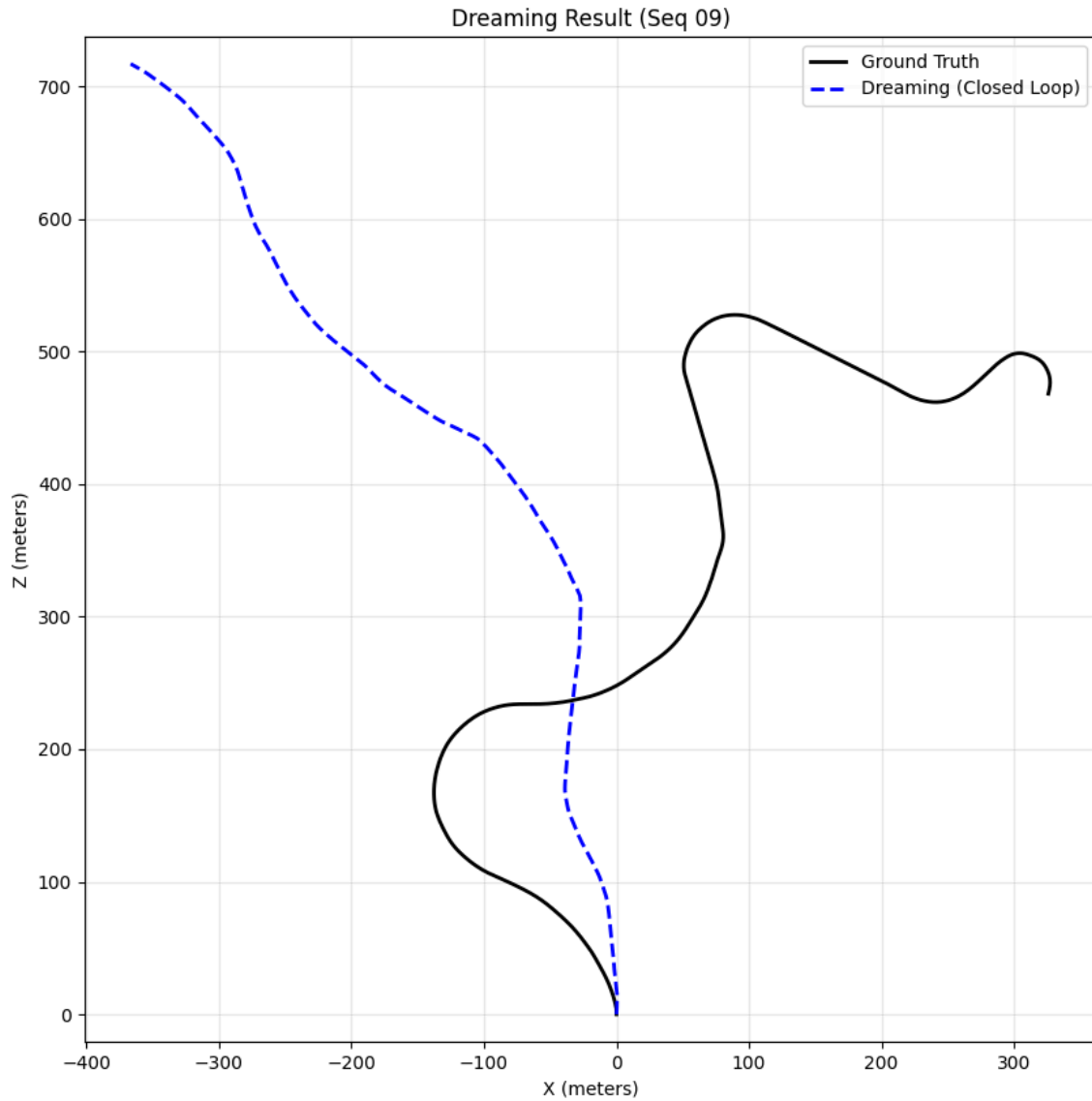


⚠️ Pretty accurate, realistic physics, but not good

6. The Power of "Dreaming"

Closed-Loop Simulation

(Model feeds its own predictions back forever)



⚠ Drifts a lot from start.

Not stable enough for 5-10 second planning horizons.

Conclusion

We built a functional World Model:

1. **Compresses** vision (VAE).
2. **Remembers** context (RNN).
3. **Simulates** uncertain futures (MDN).

Next Step: Model-Based RL

Train a driving agent entirely inside this safe simulation.